Statement of Research Interest

My research interest lies at the intersection of machine learning, natural language processing, and practical AI systems. During my Bachelor of Engineering in Computer Science and Design at Atria Institute of Technology, I became particularly interested in how AI models can be applied to real-world problems while remaining understandable and useful to people. My academic and project experience has given me exposure to machine learning, AI, databases, and software development, and I now want to build on this foundation through research. I am especially interested in NLP, semantic understanding, explainable AI, and efficient AI systems.

My main research project has been a semantic plagiarism detection system, which was published in IEEE Xplore as “AI-Powered Plagiarism Detection: A Web-Based System for Text Integrity” in 2025. The system uses Sentence-BERT embeddings and cosine similarity to identify semantic similarity between texts, while lexical overlap is combined with the semantic score to reduce false positives. I also integrated Google Custom Search API functionality into the detection workflow. Working on this project made me interested in the gap between simple word matching and genuine language understanding. It also taught me that model performance is only one part of an AI system; evaluation, reliability, and how results are presented to users are equally important.

I have also worked on EcoScan, a sustainability intelligence project using Gemini for zero-shot product recognition. I designed a weighted Sustainable Score Engine and an end-to-end workflow for real-time vision inference using FastAPI. In another project, I developed a real-time hearing-aid application using digital signal processing, including 512-point FFT analysis, spectral subtraction for noise cancellation, and dynamic parametric equalization. Although this project is not purely an ML project, it strengthened my interest in signal processing, efficient inference, and building AI-related systems under practical latency constraints.

At this stage, I do not have a fixed PhD project proposal. Instead, I would like to use the Research Symposium to understand current research directions and identify a focused question that I could develop into future postgraduate research. I am particularly interested in learning how researchers formulate strong research questions, design meaningful experiments, evaluate AI systems rigorously, and identify limitations in existing methods. I would value discussions with researchers and other students because they would help me move from project-based development toward a more research-oriented way of thinking.

My motivation for attending the Research Symposium is not simply to add another credential to my profile. I want exposure to how current AI research is conducted and to understand where my technical interests could develop into a meaningful research direction. I hope the symposium will help me learn from ongoing work, receive different perspectives, and develop a clearer foundation for pursuing future research in machine learning and AI.